

Fleet Management – UI/UX Case Study

Title:

Fleet Management Mobile App

Objective:

Design a mobile app that allows fleet managers to:

- Track multiple vehicles in real time
- Monitor vehicle status
- Identify breakdowns or delays quickly
- Take immediate action when issues occur

The focus was on **clarity, usability, and logical flow**, not complex features.

1. Empathize (User Assumptions & Pain Points)

Target User:

Fleet Manager in a logistics company

Assumptions:

- Manages multiple trucks at the same time
- Needs real-time updates
- Works in a fast-paced environment
- Makes quick decisions under pressure

Pain Points Identified:

- Hard to track many vehicles simultaneously
- Difficulty identifying urgent breakdowns
- Delays in responding to issues
- Information overload in complex dashboards
- Switching between multiple systems

User Needs:

- Instant fleet overview
- Clear issue prioritization
- Real-time location tracking

- ETA visibility
- Quick action buttons

Design Goal from Empathy Stage:
Make the app scannable within 5–10 seconds.

2. Define (Clear Problem Statement)

Problem Statement:

Fleet managers need a simple and clear mobile interface to monitor multiple trucks in real time, identify delays or breakdowns quickly, and take immediate action without navigating through complex dashboards.

Key Design Focus:

- Clarity over complexity
- Logical workflow
- Fast decision-making support

3. Ideate (Ideas Explored & Final Direction)

Option 1: List-First Design

- Main screen shows all trucks in a list
- Tap to open details

Problem:

- Hard to visually track location
- Not intuitive for real-time monitoring

Option 2: Dashboard + Map-Based Tracking (Selected)

Structure:

- Overview Dashboard for status summary
- Map for real-time tracking
- Alerts for issues
- Profile for settings

Why Selected:

- Managers scan status first

- Then visually locate trucks
- Then take action

Final Concept:

Scan → Locate → Understand → Act

4. Prototype (Screens & Flow Decisions)

High-Fidelity Screens Designed:

1. Welcome Screen

- App name
- Tagline: Track. Act. Deliver.
- CTA: Get Started

Purpose:

Simple entry point.

2. Dashboard (Overview)

- Total vehicles count
- Status cards (On Route, Delayed, Breakdown, Maintenance)
- Active Issues list

UX Decision:

Active Issues = Delayed + Breakdown

Helps managers instantly identify problems.

3. Fleet Map

- Real-time map
- Vehicle markers
- Filter chips (All / Issues / On Route)
- Bottom sheet vehicle preview

UX Decision:

Visual tracking improves faster decision-making than list-only view.

4. Vehicle Details

- Driver name

- Live location timestamp
- Destination
- ETA
- Route progress indicator
- Fuel level
- Engine status

UX Decision:

Combine operational data + vehicle health in one screen.

5. Alerts / Issues & Actions

- Active issue count
- Severity indicators (High / Low)
- Resolve button
- Contact button

UX Decision:

Prioritize critical problems visually.

6. Profile Page

- Manager information
- Contact details
- Fleet overview preview
- Logout option

UX Decision:

Separate operational tasks from account management to reduce cognitive load.

➤ Information Architecture

Bottom Navigation:

- Overview
- Map
- Alerts
- Profile

Vehicle Details is a drill-down screen.

➤ User Flow

Welcome

→ Dashboard

→ Map

→ Vehicle Details

→ Alerts

→ Action (Resolve / Contact)

→ Profile

Logic:

Scan → Locate → Understand → Act → Manage

5. Test (What I Would Validate or Improve)

If tested with real fleet managers, I would validate:

Usability Testing:

- Can users identify issues within 5 seconds?
- Is the status hierarchy clear?
- Are action buttons easy to find?

Map Interaction:

- Is filtering intuitive?
- Are markers clearly visible?

Vehicle Details:

- Is ETA understandable?
- Is route progress clear?

Alerts:

- Does severity help prioritize?
- Is issue count accurate?

Profile:

- Is logout clear?
- Are settings easy to manage?

➤ **Possible Improvements After Testing:**

- Add search bar for trucks
- Add sorting by severity
- Add notification badge counter
- Add push notification preview
- Add dark mode toggle